

Build DiD Monthly Panel

Construct DiD Panel

```
library(dplyr)
library(stringr)
library(readr)
library(haven)
library(lubridate)

sum_or_na <- function(x) {
  if (all(is.na(x))) {
    NA_real_
  } else {
    sum(x, na.rm = TRUE)
  }
}

clean_digit_code <- function(x, width = NULL) {
  out <- str_replace_all(as.character(x), "[^0-9]", "")
  if (!is.null(width)) {
    out <- str_pad(out, width = width, side = "left", pad = "0")
  }
  out
}

pick_first_existing <- function(candidates, names_vec) {
  hit <- candidates[candidates %in% names_vec]
  if (length(hit) == 0) NA_character_ else hit[[1]]
}

panel_candidates <- c(
  "../results/data/panel_hts10_monthly.csv",
```

```

    "../.../transformed_data/panel_hts10_monthly.csv",
    "../.../panel_hts10_monthly.csv"
  )
  panel_path <- panel_candidates[file.exists(panel_candidates)][1]
  if (is.na(panel_path)) {
    stop("Could not find panel_hts10_monthly.csv in expected paths.")
  }

  tar_candidates <- c(
    "../.../transformed_data/data_files_aer_2013/tar_val.dta",
    "../.../tar_val.dta"
  )
  tar_path <- tar_candidates[file.exists(tar_candidates)][1]
  if (is.na(tar_path)) {
    stop("Could not find tar_val.dta in expected paths.")
  }

  cat("Using panel file:", panel_path, "\n")

```

Using panel file: ../.../results/data/panel_hts10_monthly.csv

```
cat("Using tariff file:", tar_path, "\n")
```

Using tariff file: ../.../transformed_data/data_files_aer_2013/tar_val.dta

```

panel <- read_csv(panel_path, show_col_types = FALSE) %>%
  mutate(
    Country = as.character(Country),
    hts10 = clean_digit_code(hts10, width = 10),
    date = as.Date(date),
    year = year(date),
    month_num = month(date),
    customs_value = suppressWarnings(as.numeric(customs_value)),
    first_unit_qty = suppressWarnings(as.numeric(first_unit_qty)),
    quantity_description = as.character(quantity_description)
  ) %>%
  filter(!is.na(date), !is.na(hts10), nchar(hts10) == 10)

# Restrict to China panel per design.
panel <- panel %>% filter(Country == "China")

```

```

# Drop known HTS10 codes with quantity-description switches.
problem_switch_hts10 <- c(
  "2607000020", "2711290060", "6111206050", "6111305050", "6111905050",
  "7115900530", "7115900590", "7115903000", "7115904000", "7115906000",
  "9001300000"
)

rows_before_drop <- nrow(panel)
panel <- panel %>% filter(!(hts10 %in% problem_switch_hts10))
rows_after_drop <- nrow(panel)

cat("Dropped rows from 11 problematic HTS10 codes:", rows_before_drop - rows_after_drop, "\n")

```

Dropped rows from 11 problematic HTS10 codes: 720

```

# Build HS8-month panel.
panel_hs8 <- panel %>%
  mutate(
    hs8 = str_sub(hts10, 1, 8),
    unit_label = if_else(is.na(quantity_description), "", str_trim(quantity_description))
  ) %>%
  group_by(hs8, date, year, month_num) %>%
  summarise(
    customs_value = sum_or_na(customs_value),
    unique_unit_count = n_distinct(unit_label[unit_label != ""]),
    unit_consistent = as.integer(unique_unit_count <= 1),
    quantity_description = case_when(
      unique_unit_count == 1 ~ first(unit_label[unit_label != ""]),
      TRUE ~ NA_character_
    ),
    first_unit_qty_hs8 = if_else(
      unique_unit_count <= 1,
      sum_or_na(first_unit_qty),
      NA_real_
    ),
    n_hts10_in_hs8 = n_distinct(hts10),
    .groups = "drop"
  ) %>%
  mutate(
    unit_value = if_else(
      !is.na(first_unit_qty_hs8) & first_unit_qty_hs8 > 0,

```

```

      customs_value / first_unit_qty_hs8,
      NA_real_
    )
  )

# Load tar_val and identify code + gap fields.
tar <- read_dta(tar_path) %>% as_tibble()
tar_names <- names(tar)

year_col <- pick_first_existing(c("year", "Year", "YEAR"), tar_names)
if (is.na(year_col)) stop("Could not identify year column in tar_val.dta.")

gap_col <- pick_first_existing(
  c("spread", "ntr_gap", "gap", "gap_pre", "NTR_gap"),
  tar_names
)
if (is.na(gap_col)) stop("Could not identify NTR-gap column in tar_val.dta.")

code_col <- pick_first_existing(
  c("hs10", "hts10", "hs8", "hts8", "hs"),
  tar_names
)
if (is.na(code_col)) {
  hs_like <- tar_names[str_detect(tolower(tar_names), "hs|hts|code")]
  if (length(hs_like) == 0) stop("Could not identify HS code column in tar_val.dta.")
  code_col <- hs_like[[1]]
}

value_col <- pick_first_existing(c("v", "value", "trade_value"), tar_names)

cat("tar_val fields selected -> code:", code_col, "| year:", year_col, "| gap:", gap_col, "\n")

```

tar_val fields selected -> code: hs8 | year: year | gap: spread

```

tar_1999 <- tar %>%
  transmute(
    year = suppressWarnings(as.integer(.data[[year_col]])),
    code_raw = .data[[code_col]],
    gap = suppressWarnings(as.numeric(.data[[gap_col]])),
    weight_v = if (!is.na(value_col)) suppressWarnings(as.numeric(.data[[value_col]])) else 1
  ) %>%

```

```

filter(year == 1999) %>%
mutate(
  code_digits = clean_digit_code(code_raw),
  hs8 = str_sub(str_pad(code_digits, width = 8, side = "left", pad = "0"), 1, 8)
) %>%
filter(!is.na(gap), !is.na(hs8), nchar(hs8) == 8)

# Collapse to HS8 gap_pre (weighted by v when available, otherwise mean).
gap_pre_hs8 <- tar_1999 %>%
  group_by(hs8) %>%
  summarise(
    gap_pre = if (any(!is.na(weight_v) & weight_v > 0)) {
      sum(gap * if_else(is.na(weight_v), 0, weight_v), na.rm = TRUE) /
      sum(if_else(is.na(weight_v), 0, weight_v), na.rm = TRUE)
    } else {
      mean(gap, na.rm = TRUE)
    },
    .groups = "drop"
  )

# Merge gap_pre onto HS8-month trade panel.
did_panel <- panel_hs8 %>%
  left_join(gap_pre_hs8, by = "hs8")

# DiD indicators.
policy_date <- as.Date("2002-01-01")
median_gap <- median(did_panel$gap_pre, na.rm = TRUE)

did_panel <- did_panel %>%
  mutate(
    post = if_else(date >= policy_date, 1L, 0L),
    highgap = if_else(!is.na(gap_pre) & gap_pre >= median_gap, 1L, 0L, missing = NA_integer_),
    treated = if_else(!is.na(highgap), highgap * post, NA_integer_),
    event_time = (year(date) - year(policy_date)) * 12 + (month(date) - month(policy_date)),
    ln_customs_value = log1p(customs_value),
    ln_first_unit_qty = if_else(!is.na(first_unit_qty_hs8), log1p(first_unit_qty_hs8), NA_real_),
    ln_unit_value = if_else(!is.na(unit_value) & unit_value > 0, log(unit_value), NA_real_)
  ) %>%
  arrange(hs8, date)

# Duplicate-key check (hs8, date).
dup_count <- did_panel %>%

```

```

count(hs8, date, name = "n") %>%
filter(n > 1) %>%
nrow()

# Save output.
results_data_dir <- "../..//results/data"
dir.create(results_data_dir, recursive = TRUE, showWarnings = FALSE)
did_panel_out <- file.path(results_data_dir, "did_panel_monthly.csv")

write_csv(did_panel, did_panel_out, na = "")
cat("\nSaved", did_panel_out, "\n")

```

Saved ../..//results/data/did_panel_monthly.csv

```

# Diagnostics.
cat("\nDiagnostics\n")

```

Diagnostics

```
cat("Rows:", nrow(did_panel), "\n")
```

Rows: 813528

```
cat("Unique hs8:", n_distinct(did_panel$hs8), "\n")
```

Unique hs8: 9711

```
cat("Date range:", as.character(min(did_panel$date, na.rm = TRUE)), "to", as.character(max(did_panel$date, na.rm = TRUE)), "\n")
```

Date range: 1996-01-01 to 2005-12-01

```
cat("Share unit_consistent==0:", mean(did_panel$unit_consistent == 0, na.rm = TRUE), "\n")
```

Share unit_consistent==0: 0.01809895

```
cat("Duplicate keys (hs8, date):", dup_count, "\n")
```

Duplicate keys (hs8, date): 0

```
cat("\nSummary of gap_pre:\n")
```

Summary of gap_pre:

```
print(summary(did_panel$gap_pre))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
0.000	0.213	0.349	0.350	0.450	3.068	47004

```
cat("\nHighgap counts:\n")
```

Highgap counts:

```
print(table(did_panel$highgap, useNA = "ifany"))
```

0	1
430176	383352

```
cat("\nSummary of customs_value (non-missing):\n")
```

Summary of customs_value (non-missing):

```
print(summary(did_panel$customs_value[!is.na(did_panel$customs_value)]))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.000e+00	0.000e+00	2.781e+04	1.453e+06	2.999e+05	1.082e+09

```
cat("\nSummary of unit_value (non-missing):\n")
```

Summary of unit_value (non-missing):

```
print(summary(did_panel$unit_value[!is.na(did_panel$unit_value)]))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0	1.7	5.3	737.1	29.0	9950000.0

Dropped Unit-Switch HTS10 Codes

Removed codes due to quantity-description switches (no conversion imposed):

- 2607000020: kilograms (1996) -> component kilograms (2002)
- 2711290060: no units collected (1996-2001) -> barrels (2002-2005)
- 6111206050: dozens (2004) -> dozen pairs (2005)
- 6111305050: dozens (2004) -> dozen pairs (2005)
- 6111905050: dozens (2004) -> dozen pairs (2005)
- 7115900530: no units collected (1999-2001) -> grams (2003, 2005 observed)
- 7115900590: no units collected (2001) -> grams (2004-2005)
- 7115903000: no units collected (1996-2001) -> grams (2002, 2004-2005 observed)
- 7115904000: no units collected (1996-2001) -> grams (2002-2005)
- 7115906000: no units collected (1997-2001) -> grams (2002-2005)
- 9001300000: pairs (1999-2004 observed) -> number (2005)